

CDAC Mumbai

Lab Assignment 3

Problem 1: Sum of Two Numbers (Using a Method)

Problem Statement:

Write a Java program that includes a method to calculate the sum of two numbers.

1. Create a method `sumOfTwoNumbers()` that takes two integers as parameters, calculates their sum, and returns the result.
2. In the main method, use the `Scanner` class to prompt the user to enter two integers.
3. Pass the user inputs to the `sumOfTwoNumbers()` method and print the sum.

Sample Input:

Enter first number: 15

Enter second number: 25

Expected Output:

The sum of 15 and 25 is 40.

Java Code:

```
import java.util.Scanner;

public class Sum {
    public static void main(String[] args) {

        Scanner sc =new Scanner(System.in);
        System.out.print("Enter first number: ");
        int a =sc.nextInt();

        System.out.print("Enter second number: ");
        int b = sc.nextInt();

        int sum =a+b;
        System.out.println("Sum = "+ sum);
    }
}
```

Output:

```
PS F:\Java Code\Logic Building\Assignment 2> javac Sum.java
PS F:\Java Code\Logic Building\Assignment 2> java Sum
Enter first number: 15
Enter second number: 25
Sum = 40
PS F:\Java Code\Logic Building\Assignment 2>
```

➤ Output with different values as well:

1. (Enter first number: 27 & Enter second number: 26)

```
PS F:\Java Code\Logic Building\Assignment 2> javac Sum.java
PS F:\Java Code\Logic Building\Assignment 2> java Sum
Enter first number: 27
Enter second number: 26
Sum = 53
PS F:\Java Code\Logic Building\Assignment 2>
```

Problem 2: Simple Age Checker (Using a Method)

Problem Statement:

Write a Java program that includes a method to check the age category. ‘

1. Create a method checkAgeCategory() that takes an integer (age) as a parameter and prints whether the user is a minor, adult, or senior citizen.
2. In the main method, use the Scanner class to prompt the user to enter their age.
3. Pass the user's age to the checkAgeCategory() method.

Sample Input:

Enter your age: 30

Expected Output:

You are an adult.

Java Code:

```
import java.util.Scanner;

public class Age {
    public static void main(String[] args) {

        Scanner sc =new Scanner(System.in);

        System.out.print("Enter your age: ");
        int age =sc.nextInt();

        if (age <18) {
            System.out.println("You are a Minor");
        } else if (age<60) {
            System.out.println("You are an Adult");
        }
        else {
            System.out.println("Your are a Senior Citizen");
        }
    }
}
```

Output:

```
PS F:\Java Code\Logic Building\Assignment 2> javac Age.java
```

```
PS F:\Java Code\Logic Building\Assignment 2> java Age
```

```
Enter your age: 65
```

```
Your are a Senior Citizen
```

```
PS F:\Java Code\Logic Building\Assignment 2> java Age
```

```
Enter your age: 19
```

```
You are an Adult
```

```
PS F:\Java Code\Logic Building\Assignment 2> java Age
```

```
Enter your age: 15
```

```
You are a Minor
```

```
PS F:\Java Code\Logic Building\Assignment 2>
```

Problem 3: Print Even Numbers (Using while Loop)

Problem Statement:

Write a Java program that prints all even numbers between 1 and 50 using a while loop.

1. Create a method printEvenNumbers() that prints all even numbers from 1 to 50.
2. Use a while loop to iterate from 1 to 50 and print the even numbers.

Sample Output:

2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40 42 44 46 48 50

Java Code:

```
public class EvenNum {  
    public static void main(String[] args) {  
  
        int i =1;  
  
        while (i<=50){  
            if (i % 2== 0) {  
                System.out.print(i +" ");  
            }  
            i++;  
        }  
    }  
}
```

Output:

PS F:\Java Code\Logic Building\Assignment 2> javac EvenNum.java

PS F:\Java Code\Logic Building\Assignment 2> java EvenNum

2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40 42 44 46 48 50

Problem 4: User Input for Positive Numbers (Using do-while Loop)

Problem Statement:

Write a Java program that repeatedly asks the user to enter a positive number. ‘

1. Create a method askForPositiveNumber() that uses a do-while loop to ask the user for a number until they enter a positive number.
2. Use the Scanner class to take the user's input.
3. Once a positive number is entered, the program should display the number.

Sample Input:

Enter a positive number: -5

Enter a positive number: 0

Enter a positive number: 8

Expected Output:

You entered a positive number: 8

Java Code:

```
import java.util.Scanner;

public class PositiveNumber {
    public static void main(String[] args) {

        Scanner sc =new Scanner(System.in);
        int number;

        do {
            System.out.print("Enter the positive number: ");
            number =sc.nextInt();
        }
        while (number <=0);
        System.out.println("You entered positive number: " +number);
    }
}
```

Output:

PS F:\Java Code\Logic Building\Assignment 2> javac PositiveNumber.java

PS F:\Java Code\Logic Building\Assignment 2> java PositiveNumber.java

Enter the positive number: -5

Enter the positive number: 0

Enter the positive number: 8

You entered positive number: 8

PS F:\Java Code\Logic Building\Assignment 2> java PositiveNumber.java

Enter the positive number: -7

Enter the positive number: -5

Enter the positive number: -4

Enter the positive number: 0

Enter the positive number: 16

You entered positive number: 16

PS F:\Java Code\Logic Building\Assignment 2>

Problem 5: Print Multiplication Table (Using for Loop)

Problem Statement:

Write a Java program that prints the multiplication table for a given number (e.g., number 5) using a for loop. The program should:

1. Create a method printMultiplicationTable() that takes a number as a parameter and prints its multiplication table from 1 to 10.
2. Use a for loop to iterate through numbers 1 to 10 and print the multiplication results.

Sample Input:

Enter a number: 5

Expected Output:

5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50

Java Code:

```
import java.util.Scanner;

public class MulTable {
    public static void main(String[] args) {

        Scanner sc =new Scanner(System.in);

        System.out.print("Enter the positive number: ");
        int number = sc.nextInt();

        for (int i =1; i<=10; i++) {
            System.out.println(number + "x" + i + "=" + number*i);
        }
    }
}
```


Output:

PS F:\Java Code\Logic Building\Assignment 2> javac MulTable.java

PS F:\Java Code\Logic Building\Assignment 2> java MulTable

Enter the positive number: 5

5x1=5

5x2=10

5x3=15

5x4=20

5x5=25

5x6=30

5x7=35

5x8=40

5x9=45

5x10=50

PS F:\Java Code\Logic Building\Assignment 2>

Problem 6: Calculate the Sum of Numbers from 1 to N (Using for Loop)

Problem Statement:

Write a Java program that calculates the sum of all integers from 1 to N (where N is a positive integer) using a for loop. The program should:

1. Create a method calculateSum() that takes a number N and calculates the sum of all integers from 1 to N.
2. Use a for loop to iterate through all integers from 1 to N and add them up.

Sample Input:

Enter a number: 5

Expected Output:

The sum of numbers from 1 to 5 is: 15

Java Code:

```
import java.util.Scanner;

public class SumN {
    public static void main(String[] args) {

        Scanner sc=new Scanner(System.in);
        System.out.print("Enter number: ");
        int n= sc.nextInt();

        int sum =0;

        for (int i= 1; i<=n; i++) {
            sum =sum + i;
        }
        System.out.println("The Sum of numbers from 1 to 5 is "+ sum);
    }
}
```

Output:

PS F:\Java Code\Logic Building\Assignment 2> javac SumN.java

PS F:\Java Code\Logic Building\Assignment 2> java SumN

Enter number: 5

The Sum of numbers from 1 to 5 is 15

PS F:\Java Code\Logic Building\Assignment 2>